

Fanyou Wu | Applied Scientist II & Tech Leader @ Amazon

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Technical Experience

Applied Scientist II <i>Amazon, PXT Central Science</i>	Jan 2023 — Present Seattle, WA
• 2026: build an internal HR chatbot combining retrieval-augmented generation (RAG) with an LLM agent to answer policy questions and automate multi-step HR workflows. • 2025: develop an audio role-playing agent that simulates realistic employee conversations to coach HR professionals on handling difficult discussions. • 2023 – 2024: build a transformer-based job recommendation system on large-scale click-stream data, improving candidate-to-role matching across Amazon’s hiring funnel. • Throughout: research LLMs on dialog generation, hallucination, fairness, and role playing, with results published at ACL, EMNLP, and COLM.	
Postdoc Researcher <i>Purdue University, Department of Forestry and Natural Resources</i>	Jan 2022 — Dec 2022 West Lafayette, IN
• Develop iOS App with ARKit and CoreML to estimate tree diameter, height, and species. • Use the semantic segmentation method with PyTorch to estimate tree growth ring edges images with extremely noisy.	
Machine Learning Engineer <i>Envision Digital</i>	Aug 2017 — Dec 2017 Shanghai, China
• Use LightGBM to forecast time-series electricity demand on Hadoop/Hive-based IoT platforms.	

Education

Ph.D in Forestry , <i>Purdue University</i>	Jan 2018 — Dec 2021
M.S. in Wood Material , <i>University of Eastern Finland</i>	Sept 2015 — Apr 2017
B.E in Wood Material , <i>Nanjing Forestry University</i>	Sept 2011 — Jun 2015
Exchange Student in Wood Products Processing , <i>University of British Columbia</i>	Aug 2013 — May 2014

Selected Competition Experience (Full List at wufanyou.github.io/competition)

1st KDD Cup 2024	Meta Comprehensive RAG Benchmark (CRAG): category winner in two tracks.	\$1000
2nd KDD Cup 2022	Amazon e-commerce search (Tasks 2 & 3): NLP retrieval and ranking models.	\$4000
2nd CVPR 2021	Agriculture-Vision Challenge: farmland semantic segmentation with distributed training and model ensemble.	\$4200
1st KDD Cup 2020	RL Competition (Reposition): vehicle repositioning via Deep-Q Learning on a self-built Julia simulator.	\$8000
2nd NeurIPS 2020	Traffic4cast: deep spatiotemporal models for traffic-map movie prediction.	\$5000
1st IJCAI-19	Alibaba Adversarial AI Challenge (Target Attack): adversarial attack via a modified Projected Gradient Descent.	\$5000
4th KDD Cup 2019	Regular ML Track: Baidu context-aware multi-modal transportation recommendation.	\$1000
2nd JDD 2018	Global Digitalization Challenge (population dynamics census & prediction): 1st in NA region, 2nd in the final.	\$23000

Selected Recent Publications (Left Rank = My Author Order; Full List at [Google Scholar](https://scholar.google.com/citations?user=wufanyou))

1st ACL 2024	Synthesizing Conversations from Unlabeled Documents using Automatic Response Segmentation.
1st Inf. Process. Agric. 2024	Automated Tree Ring Detection of Common Indiana Hardwood Species through Deep Learning.
1st Multimodal Transp. 2022	A Personalized Recommendation System for Multi-modal Transportation Systems.
1st arXiv 2022	Some Practice for Improving the Search Results of E-commerce.
1st Eur. J. Forest Res. 2021	Deep BarkID: A Portable Tree Bark Identification System by Knowledge Distillation.
1st Wood Sci. & Tech. 2021	Wood Identification Based on Longitudinal Section Images by using Deep Learning.
2nd Algorithms 2025	Environmental Sensitivity in AI Tree Bark Detection: Identifying Key Factors for Improving Classification Accuracy.
2nd Algorithms 2024	CentralBark Image Dataset and Tree Species Classification using Deep Learning.
2nd The Innovation 2023	Can Language Models Be Used for Real-world Urban-delivery Route Optimization?
2nd arXiv 2023	Sequence-Level Certainty Reduces Hallucination in Knowledge-Grounded Dialogue Generation.
2nd Transp. Res. Part E 2022	Deep Dispatching: A Deep RL Approach for Vehicle Dispatching on Online Ride-hailing Platform.
2nd IEEE TITS 2021	Behavior2Vector: Embedding Users’ Personalized Travel Behavior to Vector.
3rd COLM 2025	Falsereject: A Resource for Improving Contextual Safety and Mitigating Over-refusals in LLMs via Structured Reasoning.
3rd ACL 2025	Disentangling Biased Knowledge from Reasoning in Large Language Models via Machine Unlearning.
3rd arXiv 2024	HR-Agent: A Task-Oriented Dialogue (TOD) LLM Agent Tailored for HR Applications.
3rd arXiv 2024	PHAnTOM: Personality Has An Effect on Theory-of-Mind Reasoning in Large Language Models.
3rd EMNLP 2023	DeTiME: Diffusion-Enhanced Topic Modeling using Encoder-decoder based LLM.